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SIXTH SEMESTER B.Tech.(CSE)

END SEMESTER EXAMINATION May-2023

CO304 ARTIFICIAL INTELLIGENCE

Time: 3:00 Hours *Max. Marks: 40*

Note: Answer *ALL* questions.
Assume suitable missing data, if any.
CO# is course outcome(s) related to the question.

1. Answer *any TWO* of the followings

[a] Assume that you are hired by a company to evaluate the potential for using artificial intelligence (AI) techniques (not robots) to improve efficiency and reduce costs. You are given a list of tasks and are asked to identify which of the tasks can be replaced by an AI agent. The major tasks involved in the manufacturing process are: Assembly line tasks, Quality control inspections, Inventory management, Predictive maintenance, and Logistics

(i) Identify which of the tasks can be automated using AI techniques.
(ii) Evaluate the benefits and limitations of using AI to perform the identified tasks. [Maximum two sentences for each task]

[2+2] [CO1]

[b] Consider the tree shown in Fig. 1. Perform a depth-first search (DFS) of the tree. Let the start node and goal node are S and G, respectively. Also, perform a breadth-first search (BFS) of the tree. Which nodes are visited first in DFS and BFS?

[2+2] [CO2]

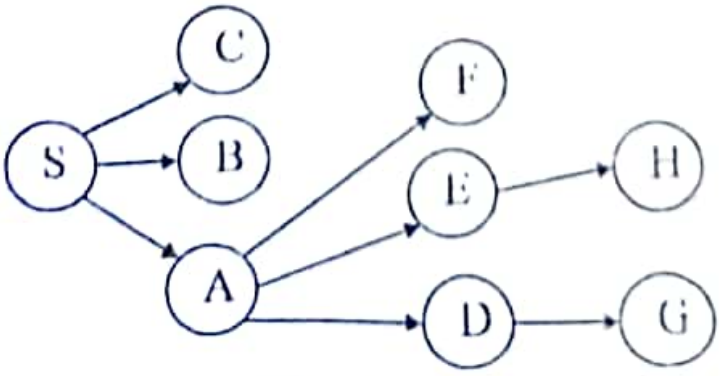

Fig. 1

[c] Suppose that the search algorithm is modified to use iterative deepening depth-first search (IDDFS). Assume that the maximum depth of the tree is 3. In what order are the nodes visited? Compare the number of nodes visited by DFS, BFS, and IDDFS. Which algorithm visits the fewest nodes in this case? [2+1+1] [CO2]

2. [a] Consider a robot that needs to navigate through a grid world from its starting position to the goal position using A* algorithm. The grid world is represented by a 4x4 grid where each cell represents a state. The robot can move up, down, left, or right in the grid world and each move has a cost of 1. Assume that the robot is initially at (0,0) and wants to reach the goal at (3,3). There are four obstacles in the grid world: (1,1), (0,2), (2,1), and (3,2). The heuristic function $h(n)$ is the Manhattan distance between node n and the goal. Apply the A* algorithm to find the optimal path from the initial state to the goal state. Show the order in which the nodes are expanded, the cost g and h values of each expanded node, and the final optimal path. [3+1] [CO1, CO2]

[b] Consider the robot navigation problem in 2[a]. Apply the uniform-cost search to find the optimal path from the initial state to the goal state. Show the order in which the nodes are expanded. Compare the performance of A* search to uniform-cost search on this problem. Which algorithm is more efficient and why? [3+1] [CO1, CO2]

3. Answer *any TWO* of the followings

[a] Consider the state of tic-tac-toe game as shown in Fig. 2. The objective of the game is to get three of your own symbols (either 'X' or '0') in a row, column, or diagonal. The symbol X represents AI player. The heuristic function for AI player is defined as below:

The heuristic function assigns to each move a score of: +10 for immediate win; +5 for blocking opponent's immediate win (in next move); +3 for creating potential winning position e.g. two X with one empty cell in a row, column, diagonal; +2 for blocking the opponent winning position; +1 for occupying the center cell.

Find the next two moves of AI player by applying Steepest-Ascent hill climbing algorithm with above heuristic function. Show each step clearly. Represent each by row and column number. Assume that opponent is a perfect player. [4] [CO1, CO2]

X		0
		X
0		

Fig. 2

[b] Consider a game tree of level 5 for two players MAX and MIN. Level 1 (MAX): A; Level 2 (MIN): B, C, D; Level 3 (MAX): E, F, G, H, I, J; Level 4 (MIN): K, L, M, N, O, P; Level 5: Q, R. Nodes B, C, & D are child nodes of A. Nodes (K, L), (M, N), (O, P), and (Q, R) are child nodes of F, G, J, and M, respectively. An evaluation function is used to assigns score to leaf nodes. The evaluation scores of leaf nodes are given in Table I. Here symbol ? represents inability of function to assign score. Is it possible to find minimax value and optimal strategy to play the game for MAX players? Also, calculate alpha and beta cut-offs if possible. [4] [CO3]

Table I

Node	E	K	L	Q	R	N	H	I	O	P
Score	2	3	?	1	?	7	6	1	?	20

[c] Explain the purpose of alpha and beta values in the alpha-beta pruning technique in minimax algorithm. Suppose you have a game tree with a depth of 4 levels. At level 3, a beta cutoff occurs during the alpha-beta pruning process. Explain the conditions that lead to a beta cutoff and how it affects the search. [2+2] [CO2]

4. [a] Consider the following propositional logic statements:

If a computer system has insufficient memory (M), then the performance degrades (D). If the performance degrades (D) or the CPU is overheating (O), then the system crashes (C). If the system crashes (C) and the hard disk fails (H), then data loss occurs (L). If data loss occurs (L), then the backup process is initiated (B). If the backup process is initiated (B) and the system is restored (S), then the downtime is minimized (X).

Prove by resolution that *if a computer system has insufficient memory (M) and the hard disk fails (H), then the downtime is minimized (X).* [4] [CO2]

[b] Consider the following set of Horn clauses and facts:

If the model is well-trained (P), then it produces accurate predictions (Q). If the dataset is large (L) and the preprocessing is done correctly (M), then the model is well-trained (P). If the regularization is applied (R) and the batch size is optimized (B), then the preprocessing is done correctly (M). If the learning rate is set appropriately (L) and the cost function is minimized (C), then regularization is applied (R). If the features are well-defined (A) and the dataset is balanced (B), then the dataset is large (L). If the features are well-defined (A) and the model produces accurate predictions (P), then the dataset is large (L). The features are well-defined (A). The dataset is balanced (B). The cost function is minimized (C).

Using forward chaining, determine whether the model produces

accurate predictions (Q) based on the given Horn clauses and facts. Use AND-OR graphs for explanation. [4] [CO2]

5. [a] Consider two deep learning models, A and B, used for detecting fraudulent transactions. Model A is 98% effective at recognizing fraudulent transactions when they occur but has a 15% false positive rate (indicating that a transaction is fraudulent, when it is not). Model B is 93% effective at recognizing fraudulent transactions, but has a 7% false positive rate. The two models use independent techniques for identifying fraud. Fraudulent transactions constitute 0.5% of all transactions. Let's say a transaction is evaluated using only one of these models, and that model flags the transaction as fraudulent. Which model's positive result is more indicative of a transaction genuinely being fraudulent? Justify your answer mathematically. [4] [CO2, CO3]

[b] Consider a Bayesian network representing the relationship between weather conditions and two variables: "*Rain*" (R) and "*Slippery Roads*" (S). The network consists of the following conditional probability Tables (CPTs): $P(Rain) = 0.3$; $P(Slippery\ Roads = \text{true} \mid Rain = \text{true}) = 0.8$; $P(Slippery\ Roads = \text{false} \mid Rain = \text{true}) = 0.2$; $P(Slippery\ Roads = \text{true} \mid Rain = \text{false}) = 0.1$; $P(Slippery\ Roads = \text{false} \mid Rain = \text{false}) = 0.9$. You encounter a situation where the roads are slippery (S = true). Based on this information and the provided Bayesian network, calculate the probability of rain (R) being true. [4] [CO2, CO3]

----Best of Luck----